Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: **B.Tech. ECE, Semester III**
Name of the Paper: **Advanced Engineering Mathematics**
Paper Code: **BSC 301**
Time: 1.5 hour

Maximum Marks: 50

Note:
(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.

Q1. (10 Marks)

a. Express the function $f(x) = \begin{cases} 1 & if\ |x| \le 1 \\ 0 & if\ |x| > 1 \end{cases}$ as a Fourier integral hence evaluate

$$\int_0^\infty \frac{\sin\lambda \cos\lambda x}{\lambda} d\lambda$$ (CO1)

OR

b. Find the Fourier transform of $f(x) = \begin{cases} 1 + \frac{x}{a} & if\ -a < x < 0 \\ 1 - \frac{x}{a} & if\ \ 0 < x < a \end{cases}$ (CO1)

Q2. (10 Marks)

a. Find the Fourier sine and cosine transform of e^{-x}. (CO1)

OR

b. Find the z-transformation of $f[n] = \begin{cases} a^n & if\ n \ge 0 \\ 0 & if\ n < 0 \end{cases}$ (CO1)

Q3. (10 Marks)

a. Find the inverse z-transform of the function $F(z) - \frac{z}{(z-2)(z-4)}$ (CO1)

OR

b. Using convolution theorem find the inverse z-transform of $F(z) = \frac{z^2}{(z\ \ 3)(z\ \ 5)}$ (CO1)

Q4. (10 Marks)

a. Determine whether the function $f(z) = 1/z$ is analytic or not. (CO2)

OR

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b. Find $\int_C \frac{2z^2+z}{(z^2-1)}dz$, where C is $|z-1|=1$. **(CO2)**

Q5. (10 Marks)

a. Show that the function $u(x,y)=\frac{1}{2}\log(x^2+y^2)$ is harmonic then find the analytic function $f(z)=u+iv$. **(CO2)**

OR

b. Evaluate $\int_C \frac{z.dz}{(9-z^2)(z+i)}$, where C is $|z|=2$. **(CO2)**